CNENGR202 Instrumentation and Control

Part 1 L3 – Nonlinear system

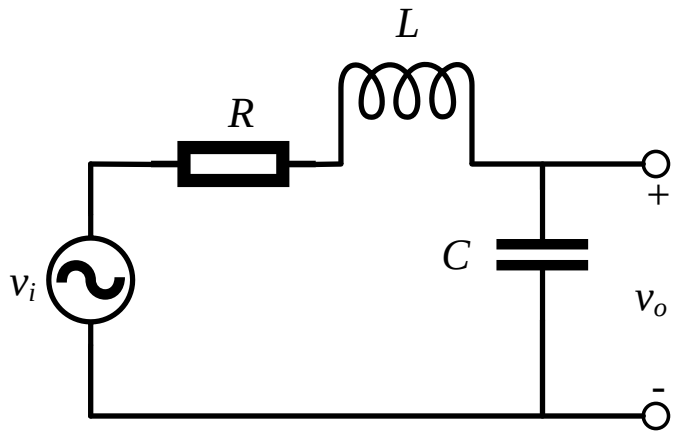
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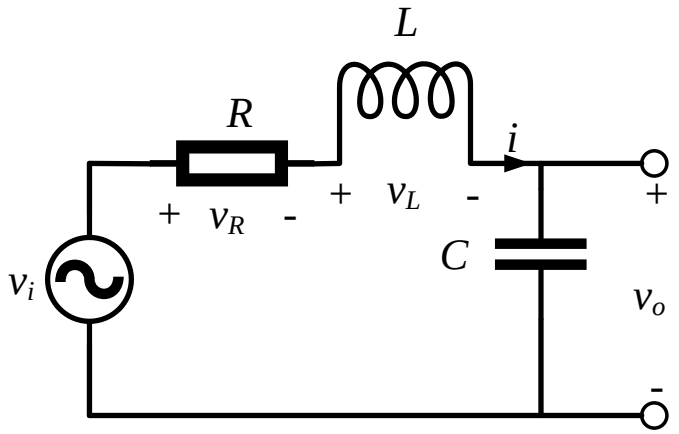
• Class question

- Determine the differential equations for the circuit shown in the right figure. Choose the capacitor voltages $v_o(t)$ as the unknowns. $v_i(t)$ is the input.



• Class question

- The capacitor voltages as the unknowns. $v_i(t)$ is the input.



$$i = C\frac{dv_o}{dt} \longrightarrow v_L = L\frac{di}{dt} = LC\frac{d^2v_o}{dt^2}$$

$$v_R = Ri = RC\frac{dv_o}{dt}$$

$$v_R + v_L + v_o = v_i \longrightarrow RC\frac{dv_o}{dt} + LC\frac{d^2v_o}{dt^2} + v_o = v_i$$

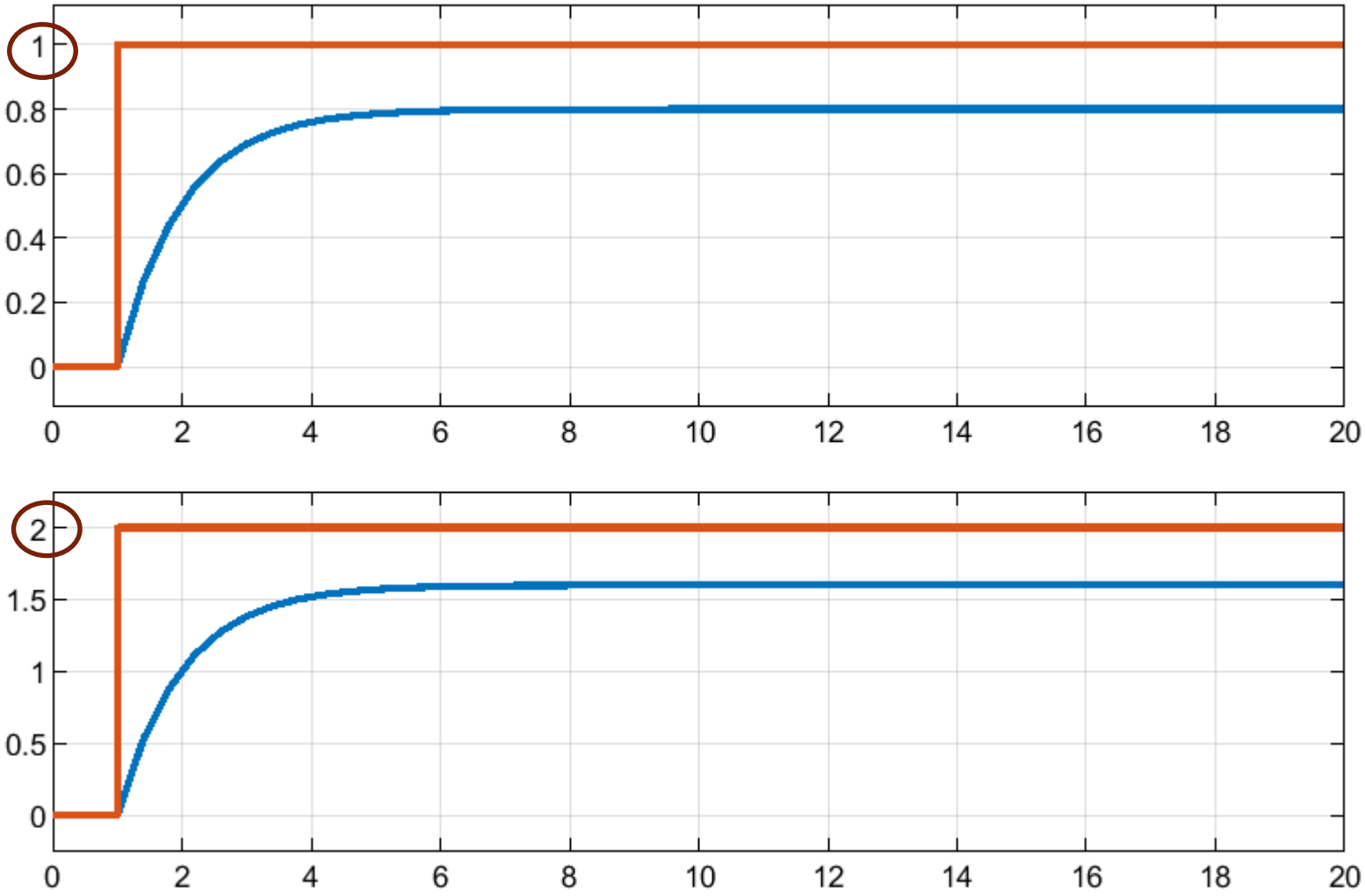
- Normalise

$$\frac{d^2v_o}{dt^2} + \frac{R}{L}\frac{dv_o}{dt} + \frac{1}{LC}v_o = \frac{1}{LC}v_i \longrightarrow \frac{d^2x}{dt^2} + 2\zeta\omega_n\frac{dx}{dt} + \omega_n^2 x = Ku$$

Outline

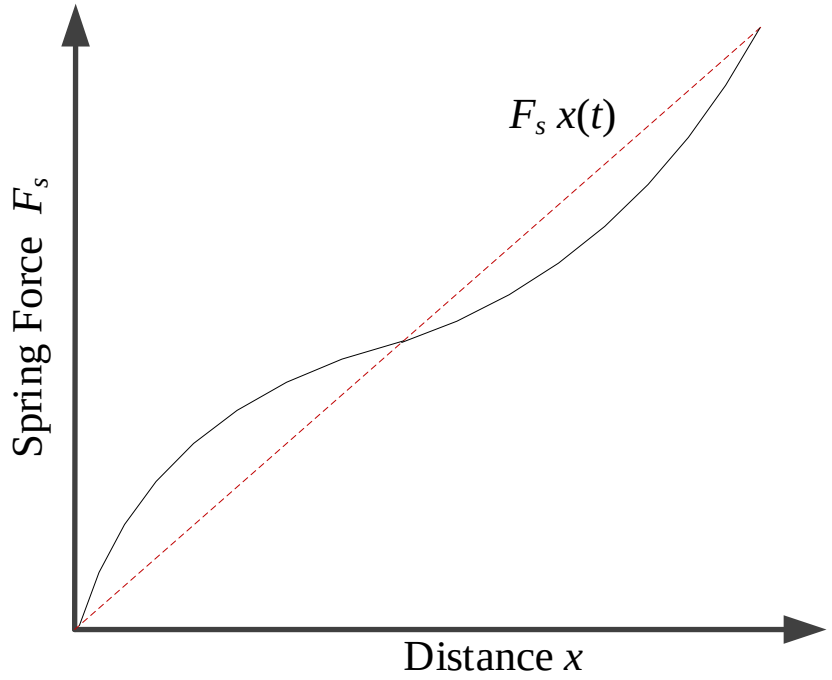
- Linear and nonlinear systems
- Continuous nonlinearity
- Discontinuous nonlinearity
- Linearisation of Continuous Non-Linearities
- Examples

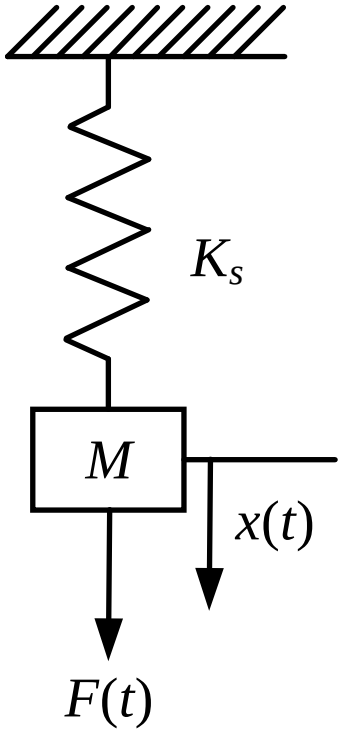
Linear and Nonlinear Systems - 1



"principle of superposition"

- Linear and Nonlinear systems - 2





- Linear and Nonlinear systems - 3

  - Linear Models

    - Obey the principle of superposition
    - Stability is an inherent property of a linear system, which is independent of the nature and amplitude of any external inputs

  - Non-linear Models

    - Can exhibit quite different response to different sized but similarly shaped inputs
    - Can exhibit complex or even chaotic behaviour

• Linear and Nonlinear systems - 3

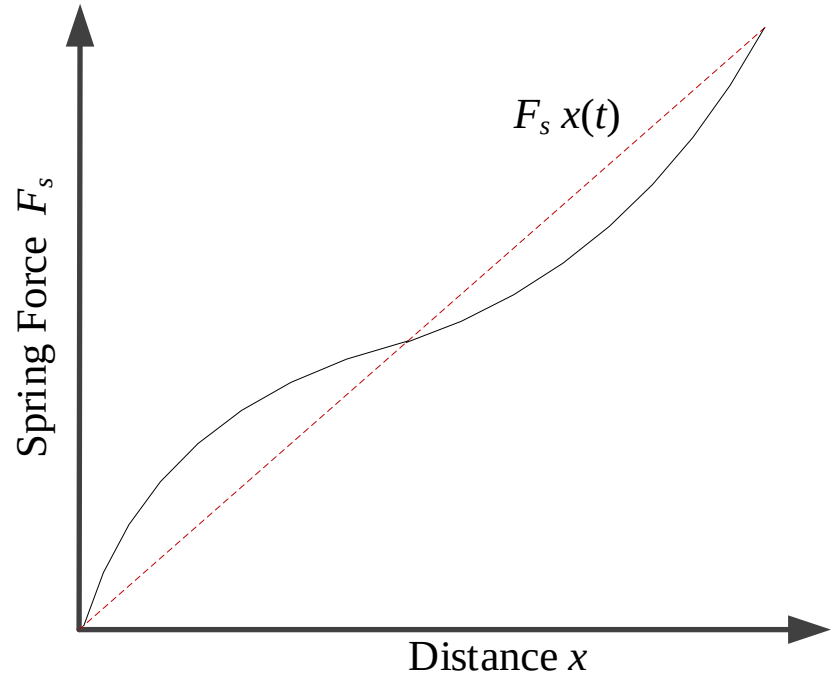
- For all the non-linear models, we need to make linearising assumptions about the system, e.g. spring force is proportional to the displacement
- However, it is not always possible to make such assumptions about the system in order to obtain a linear model
- In such cases, one approach is to linearise the non-linear model about some operating condition
- There are two type of non-linear function
  - continuous functions can be linearised
  - discountinuous functions cannot be linearised

Continuous Non-Linearity

- Consider a simple spring: normally it is assumed that spring force varies proportionally to displacement.

$$F_s = K_s \cdot x$$

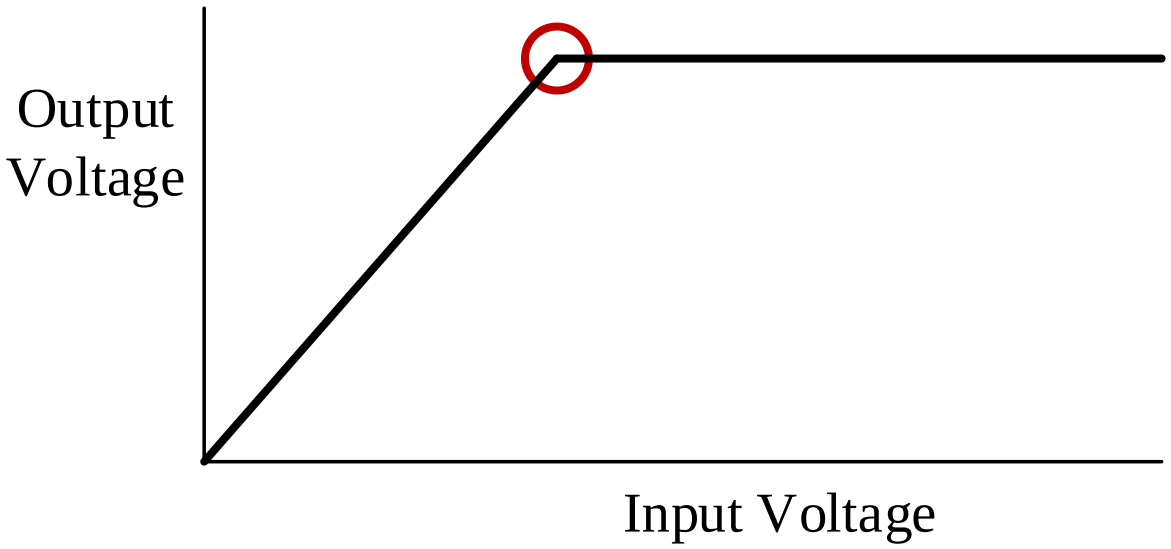
- However, in practice the relationship could look like the solid trace.



Discontinuous Non-Linearity

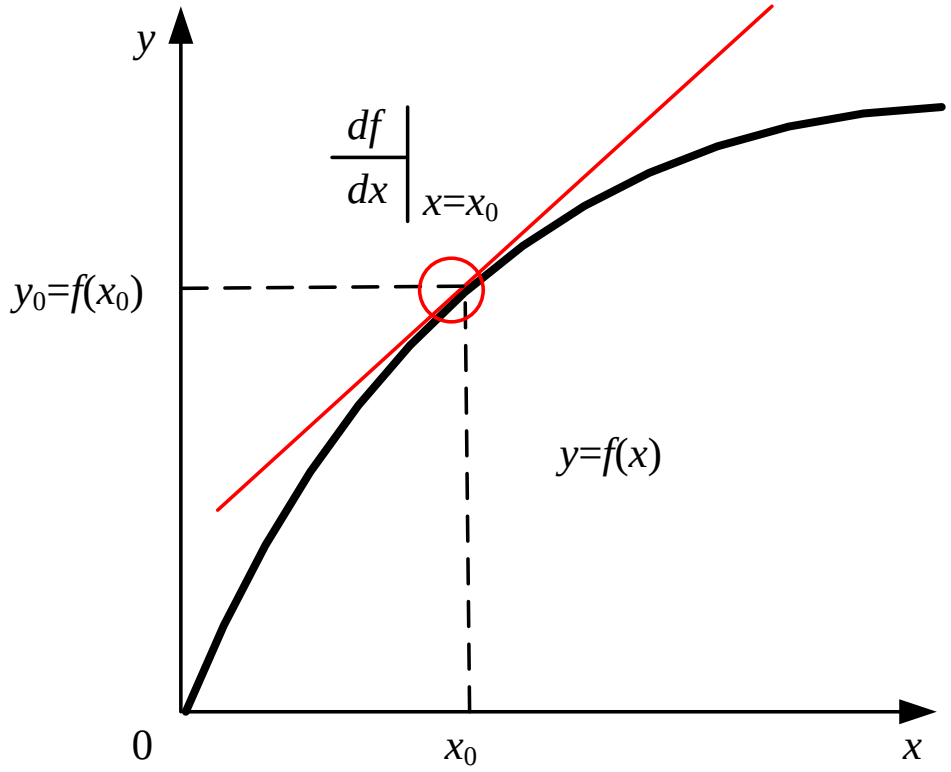
- For the operational amplifier below, the output voltage is proportional to the input voltage only up to a critical value.

- For higher voltages, the output remains constant



Linearisation of Continuous Non-Linearities -1

(x_0 , y_0) represents the operating condition of interest for this example



Linearisation of Continuous Non-Linearities -2

- The non-linear relationship can be expanded using Taylor's series where x_0 is the operating condition

$$y = f(x) = f(x_0) + \left.\frac{df}{dx}\right|_{x=x_0} (x - x_0)/1! + \left.\frac{d^2 f}{dx^2}\right|_{x=x_0} (x - x_0)^2/2! + \cdots$$

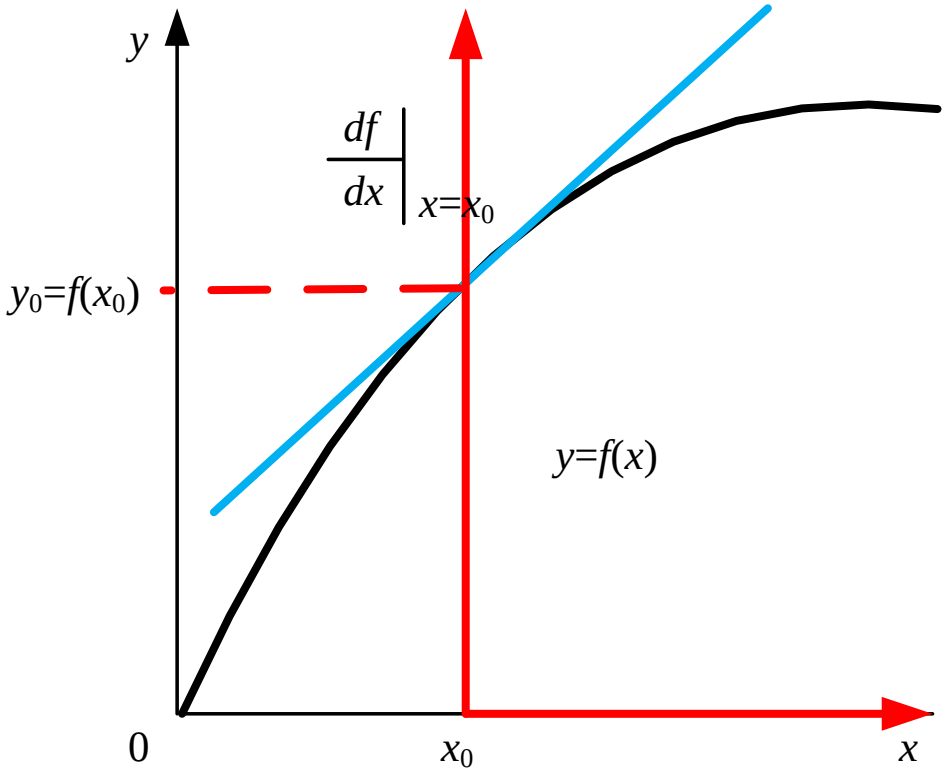
- If the deviations about the operating condition are small, then the above equation is approximated by

$$y = f(x) \approx f(x_0) + \left.\frac{df}{dx}\right|_{x=x_0} (x - x_0)$$

Linearisation of Continuous Non-Linearities -3

Intercept Slope

$$y = f(x) = f(x_0) + \left.\frac{df}{dx}\right|_{x=x_0} (x - x_0)$$



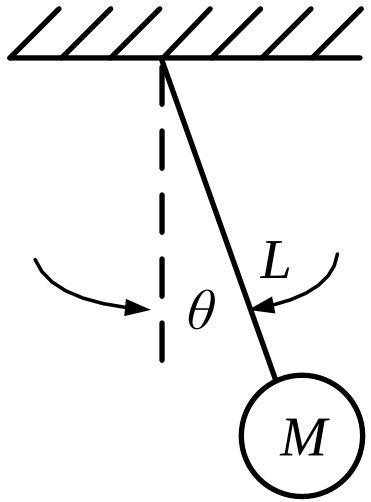
Example: Linearised Model for a Free Pendulum

$$T = MLg\sin\theta$$

$$y = f\left(x_0\right) + \left.\frac{df}{dx}\right|_{x=x_0}\left(x - x_0\right)$$

$$T = MLg\left(0 + \left.\frac{d\sin\theta}{d\theta}\right|_{\theta=0}\left(\theta - 0\right)\right)$$

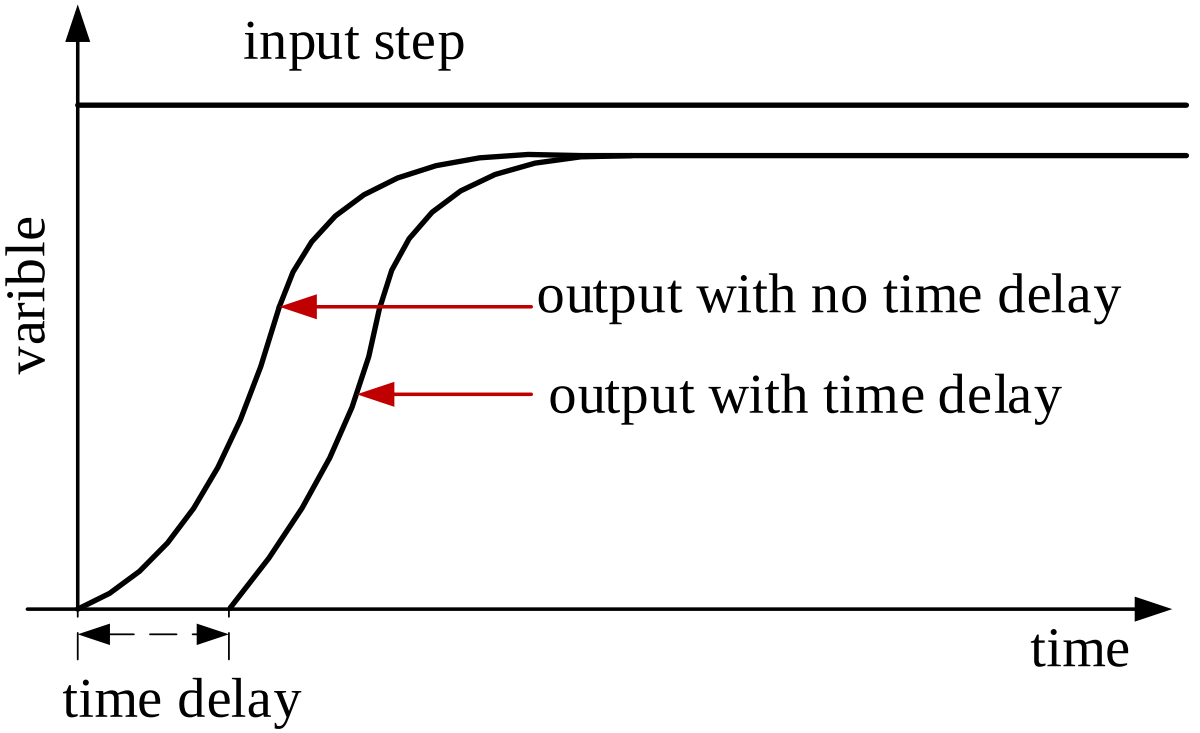
$$T = MLg\theta$$



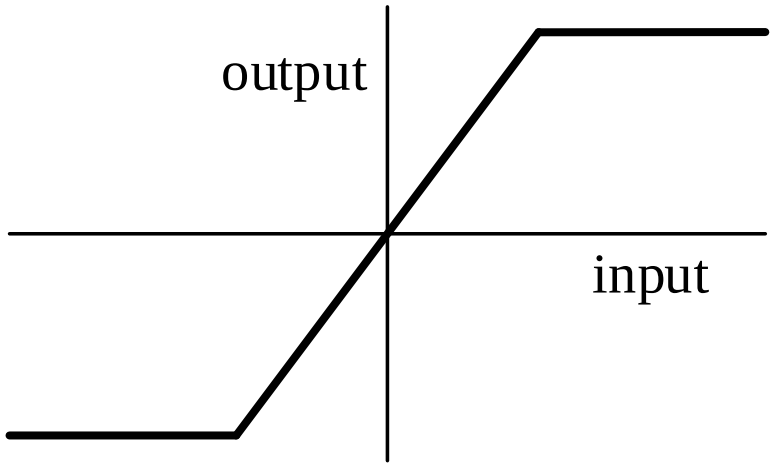
The linear model is only valid when the pendulum swings through small angles

Example : Time Delay

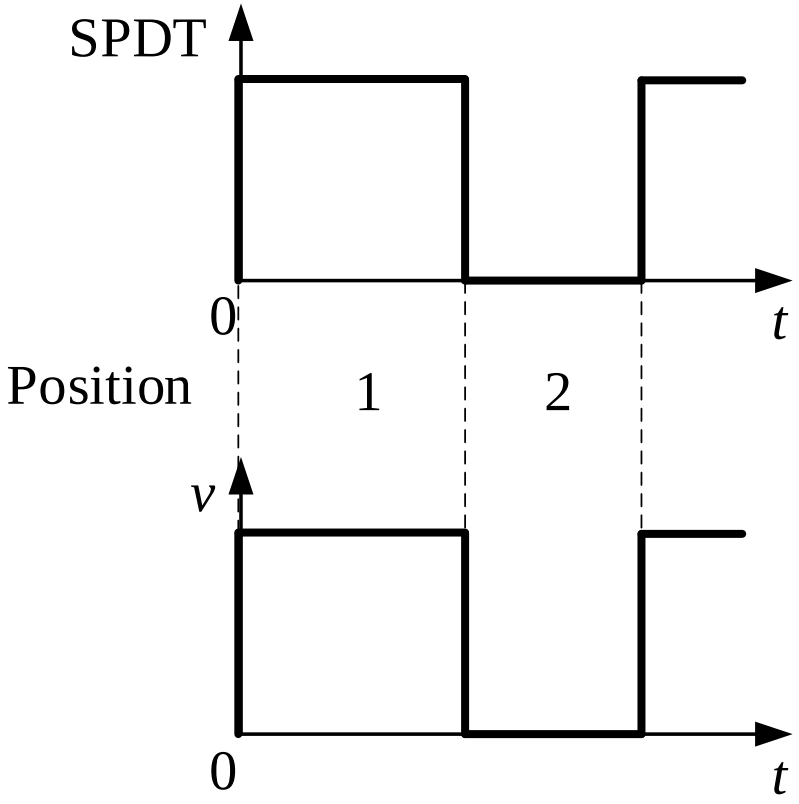
(*) Arguably not actually nonlinear!

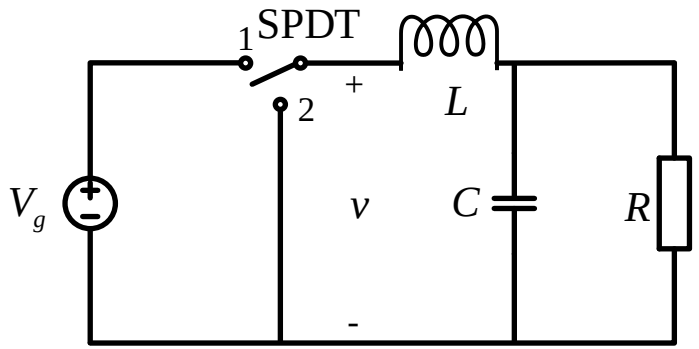


Example : Saturation

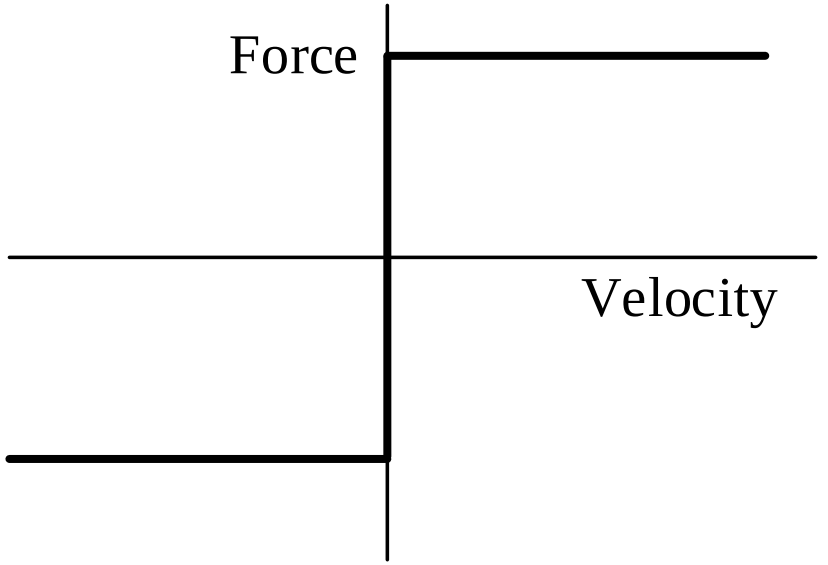


Example : SPDT

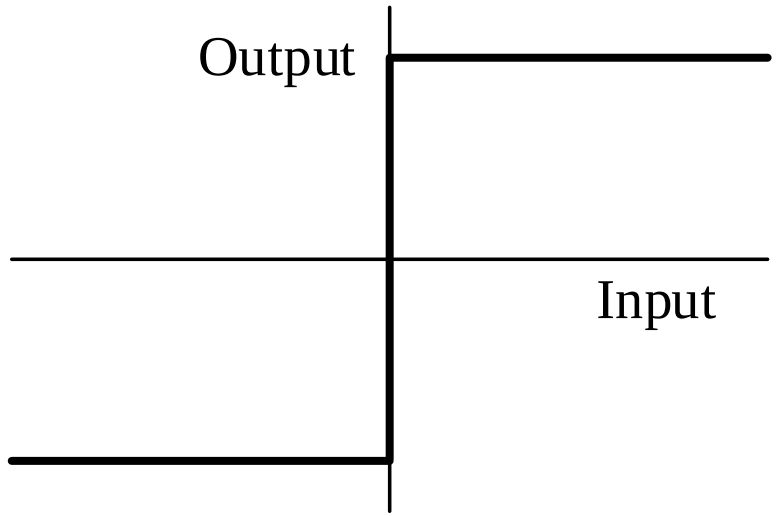




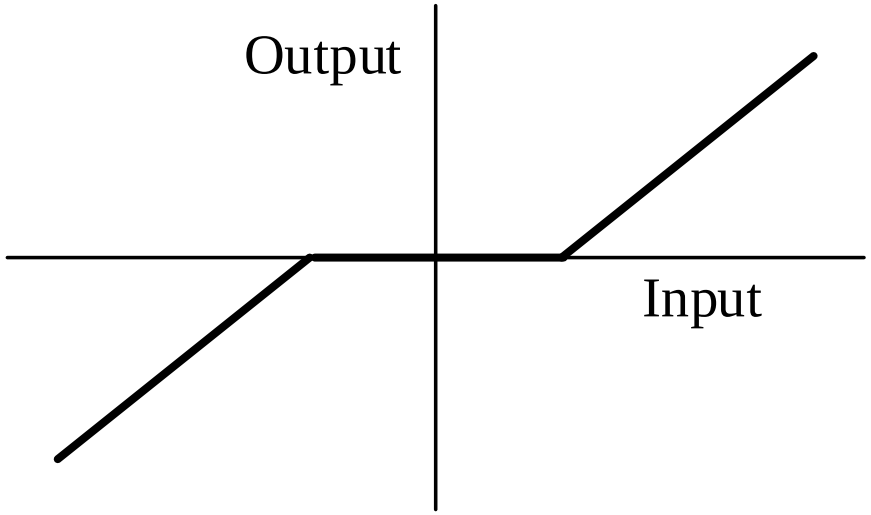
Example : Coulomb Friction



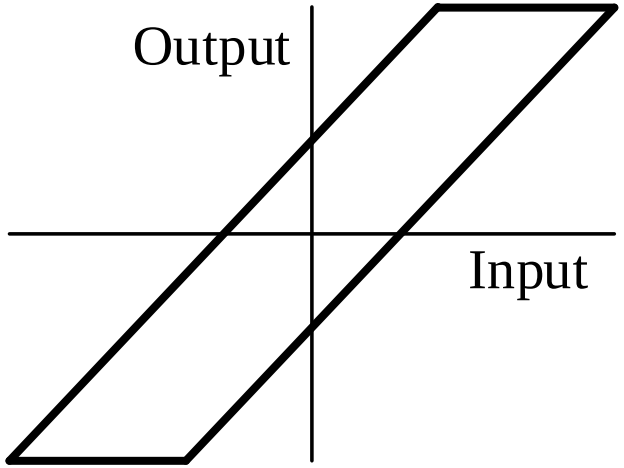
Example : Ideal Relay



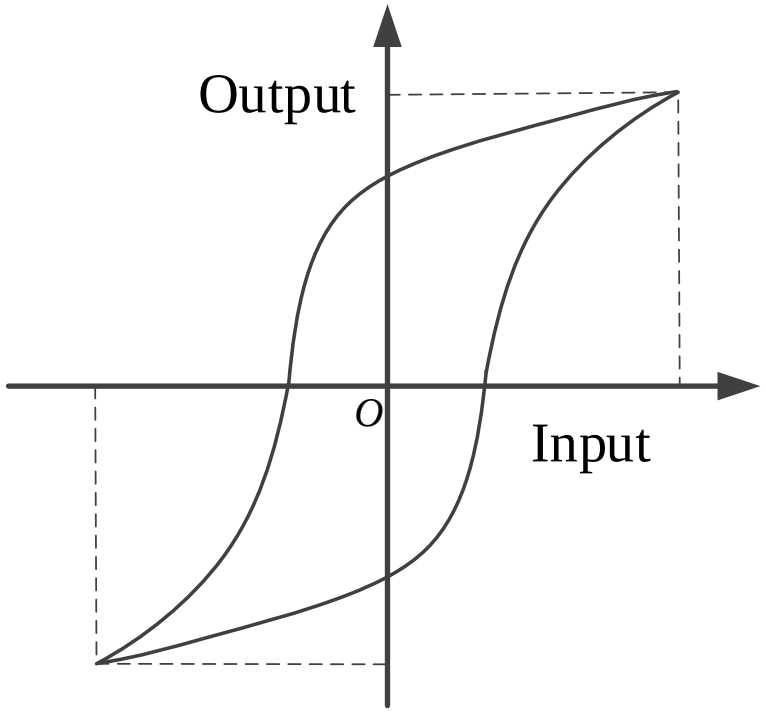
Example : Dead-Zone

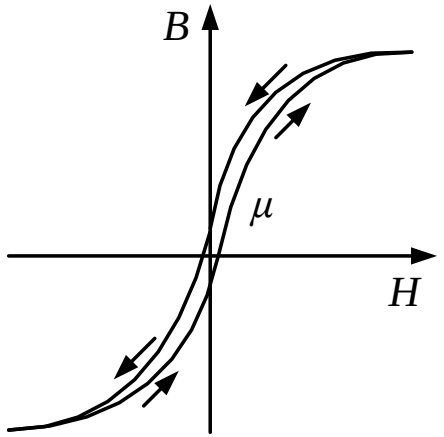


Example : Backlash



Example : Hysteresis





Summary

- By using the assumptions, physical systems can be modelled by linear time–invariant differential equations.
- The resulting linear models are only accurate when the assumptions hold.
- There are often analogies between different types of physical systems (e.g. electrical, chemical and mechanical).
- Single degree of freedom systems often fall into one of two general categories: 1st order and 2nd order.
- If these general forms can be solved, then solutions can be easily found for a wide range of physical situations
- Continuous non-linearities can be linearised.